increases the rate by a factor of 16. Find the order of the reaction with respect to x and y. what is the overall order of the reaction?

2). Find the individual and overall order of the following reaction using the given data.

$2NO(g) + Cl_2(g) \longrightarrow 2NOCl(g)$

Experiment number	Initial concentration		Initial rate
	NO	Cl_2	NOCl mol $L^{-1}s^{-1}$
1	0.1	0.1	7.8×10^{-5}
2	0.2	0.1	3.12×10^{-4}
3	0.2	0.3	9.36×10^{-4}

7.5 The integrated rate equation:

We have just learnt that the rate of change of concentration of the reactant is directly proportional to that of concentration of the reactant. For a general reaction,

$A \longrightarrow$ product

The rate law is

$$\text{Rate} = \frac{-d[A]}{dt} = k\,[A]^x$$

Where k is the rate constant, and x is the order of the reaction. The above equation is a differential equation, $\frac{-d[A]}{dt}$, so it gives the rate at any instant. However, using the above expression, we cannot answer questions such as how long will it take for a specific concentration of A to be used up in the reaction? What will be the concentration of reactant after a time ' t '?. To answer such questions, we need the integrated form of the above rate law which contains time as a variable.

7.5.1 Integrated rate law for a first order reaction

A reaction whose rate depends on the reactant concentration raised to the first power is called a first order reaction. Let us consider the following first order reaction,

$A \longrightarrow$ product

Rate law can be expressed as

$$\text{Rate} = k\,[A]^1$$

Where, k is the first order rate constant.

$$\frac{-d[A]}{dt} = k\,[A]^1$$

$$\Rightarrow \frac{-d[A]}{[A]} = k\,dt \qquad \ldots(1)$$

Integrate the above equation between the limits of time $t = 0$ and time equal to t, while the concentration varies from the initial concentration $[A_0]$ to $[A]$ at the later time.

$$\int_{[A_0]}^{[A]} \frac{-d[A]}{[A]} = k\int_0^t dt$$

$$\left(-\ln[A]\right)_{[A_0]}^{[A]} = k\,(t)_0^t$$

$$-\ln[A] - \left(-\ln[A_0]\right) = k\,(t-0)$$

$$-\ln[A] + \ln[A_0] = kt$$

$$\ln\left(\frac{[A_0]}{[A]}\right) = kt \qquad \ldots(2)$$

This equation is in natural logarithm. To convert it into usual logarithm with base 10, we have to multiply the term by 2.303.

$$2.303\log\left(\frac{[A_0]}{[A]}\right) = kt$$

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